Answer Key & Teacher Notes

Session 1 — Trapezoids by Decomposing · Session 2 — The Trapezoid Area Formula

HOW TO USE THIS LESSON'S PAGES

one lesson = two sessions

Warm-Up	About 10 minutes; use items 1–4 for a 5-minute check. Key: Unit Warm-Up Key.
Prerequisite Practice A/B	Before the lesson, for students who need the skill the session uses. Key in that folder.
Guided Notes	During the lesson. Students fill in the blanks as you teach; answers below.
Practice Form A	In class, after the lesson. Two pages. The self-check code tells students if they are on track.
Practice Form B	Same skills, new numbers: reteach, retake, small group or absent students.
Exit Ticket	Last 5 minutes. Two per page; cut in half.
Homework Packet	Sheet 1 after Session 1, Sheet 2 after Session 2. Level 1 has the same problems and answers, with a word bank, step frames and larger type for students who need language support.

SESSION 1 · TEACHER NOTES

Trapezoids by Decomposing

TALK PROMPTS

- Why must the two end triangles be the same size on an isosceles tabletop?
- How does cutting a trapezoid into simpler shapes make its area easier to find?
- How do you know the leftover part of the long base splits into two equal pieces?

PARTNER TASK

Window Frame. A trapezoid window has bases 13 cm and 25 cm, height 11 cm, and equal slanted sides. One partner finds the rectangle (143 cm²); the other finds both triangles (66 cm²). Add to get 209 cm², then switch jobs on the bus shelter roof.

WATCH FOR

- Using the whole leftover (long base − short base) as the base of EACH of two triangles instead of splitting it in half.
- Finding only the rectangle and forgetting to add the triangles.
- Using a slanted side as the height of a triangle piece.

SESSION 2 · TEACHER NOTES

The Trapezoid Area Formula

TALK PROMPTS

- Why does the formula add the two bases before anything else happens?
- How is this formula like finding the average of the two bases?
- What does the sum of the two bases tell you before you multiply by the height?

PARTNER TASK

Ramp Side. A skate-ramp side is a right trapezoid with bases 12 ft and 20 ft and height 7 ft. One partner uses the formula; the other decomposes it into a rectangle (84 ft²) and a triangle (28 ft²). Both should get 112 ft². Explain why the methods must agree.

WATCH FOR

- Forgetting the ½ (finding (b1 + b2) × h, which is the area of two trapezoids).
- Using a slanted side as the height or as a base.
- Missing base: dividing the area by the height without doubling first.

ITEM ANALYSIS

suggested DOK; judge the reasoning required, not the question label

Sheet	Standard	1	2	3	4	5	6	7	8	9	10	11
S1 · Form A	6.G.A.1	1	1	1	2	2	2	2	3	2	1	1
S1 · Form B	6.G.A.1	1	1	1	2	2	2	2	3	2	1	1
S2 · Form A	6.G.A.1	1	2	1	1	2	2	3	2	2	1	1
S2 · Form B	6.G.A.1	1	2	1	1	2	2	3	2	2	1	1

Items before the last three are Practice; then Find the Error (explain and correct) and two Test Practice items.

GUIDED NOTES

trapezoid: A **quadrilateral** with one pair of parallel sides (the bases).

height: The **perpendicular** distance between the two bases.

decompose: To **break** a shape into smaller shapes you know.

- Cut a trapezoid into a **rectangle** and one or two triangles.
- Find the area of each piece, then **add** the areas.
- Together, the triangle bases equal the **difference** of the two bases.

Example: Rectangle: 6 × 5 = **30** cm². Leftover: 14 − 6 = 8, so each triangle base is **4** cm. Both triangles: 2 × (½ × 4 × 5) = **20** cm². Answer: Area = 30 + 20 = **50** cm²

YOU TRY

1. Rectangle 6 × 4 = 24 in²; triangle ½ × 4 × 4 = 8 in²; total 32 in²

2. a. (14 − 8) ÷ 2 = 3 m b. (20 − 12) ÷ 2 = 4 cm

EXIT TICKET

1. Rectangle 9 × 12 = 108 yd²; triangle ½ × 8 × 12 = 48 yd²; total 156 yd²

2. a. 12 × 7 = 84 cm² b. 2 × (½ × 4 × 7) = 28 cm²

3. **A** (Add the areas of the pieces.)

PRACTICE FORM A

1. a. (22 − 10) ÷ 2 = 6 in b. (18 − 8) ÷ 2 = 5 m c. (20 − 12) ÷ 2 = 4 cm d. (27 − 17) ÷ 2 = 5 m

2. a. 10 × 15 = 150 in² b. (22 − 10) ÷ 2 = 6 in each; 2 × (½ × 6 × 15) = 90 in² c. 150 + 90 = 240 in²

3. Rectangle 16 × 10 = 160 ft²; triangle base 24 − 16 = 8 ft, ½ × 8 × 10 = 40 ft²; total 200 ft²

4. Rectangle 8 × 6 = 48 m²; each triangle base 5 m, both triangles 30 m²; total 78 m²

5. Rectangle 11 × 13 = 143 yd²; triangle ½ × 8 × 13 = 52 yd²; total 195 yd²

6. Tray: 19 × 16 = 304; triangles 2 × (½ × 5 × 16) = 80; tray 384 in². Mat 24 × 16 = 384 in². Yes, same area.

7. Rectangle 17 × 10 = 170 m²; each triangle base 5 m; triangles 2 × (½ × 5 × 10) = 50 m²; total 220 m²

8. a. rectangle 10 × 6 = 60 ft²; triangles 24 ft² (two of ½ × 4 × 6, or one of ½ × 8 × 6) b. 84 ft² c. 84 × 8 = $672

9. Kwame found only the rectangle and left out the two triangles. Each triangle base is (21 − 11) ÷ 2 = 5 m, so the triangles add 2 × (½ × 5 × 10) = 50 m². Area = 110 + 50 = 160 m².

10. **B** (one or two triangles)

11. **C** (46 cm²)

Self-check word **COMBINE**

PRACTICE FORM B

1. a. (20 − 12) ÷ 2 = 4 in b. (16 − 6) ÷ 2 = 5 m c. (21 − 9) ÷ 2 = 6 cm d. (23 − 15) ÷ 2 = 4 ft

2. a. 12 × 9 = 108 in² b. (20 − 12) ÷ 2 = 4 in each; 2 × (½ × 4 × 9) = 36 in² c. 108 + 36 = 144 in²

3. Rectangle 14 × 9 = 126 ft²; triangle base 22 − 14 = 8 ft, ½ × 8 × 9 = 36 ft²; total 162 ft²

4. Rectangle 6 × 8 = 48 m²; each triangle base 5 m, both triangles 40 m²; total 88 m²

5. Rectangle 9 × 14 = 126 yd²; triangle ½ × 6 × 14 = 42 yd²; total 168 yd²

6. Tray: 14 × 12 = 168; triangles 2 × (½ × 5 × 12) = 60; tray 228 in². Mat 19 × 12 = 228 in². Yes, same area.

7. Rectangle 15 × 8 = 120 m²; each triangle base 5 m; triangles 2 × (½ × 5 × 8) = 40 m²; total 160 m²

8. a. rectangle 12 × 5 = 60 ft²; triangles 20 ft² (two of ½ × 4 × 5, or one of ½ × 8 × 5) b. 80 ft² c. 80 × 9 = $720

9. Amina found only the rectangle and left out the two triangles. Each triangle base is (23 − 13) ÷ 2 = 5 m, so the triangles add 2 × (½ × 5 × 8) = 40 m². Area = 104 + 40 = 144 m².

10. **C** (height)

11. **C** (52 cm²)

Self-check word **SECTION**

GUIDED NOTES

base: One of the two **parallel** sides of a trapezoid.

height: The **perpendicular** distance between the two bases.

formula: A **rule** written with letters and numbers.

right trapezoid: A trapezoid with two **right** angles.

- Area of a trapezoid: A = ½ × (**b1 + b2**) × h.
- **Add** the bases first, then multiply by the height, then take half.
- Two copies of a trapezoid make a **parallelogram**, so the trapezoid is half of it.

Example: Add the bases: 6 + 12 = **18**. Multiply by the height: 18 × 4 = **72**. Take half: 72 ÷ 2 = **36**. Answer: A = **36** m²

YOU TRY

1. ½ × (5 + 9) × 8 = ½ × 14 × 8 = 56 cm²

2. a. ½ × 10 × 4 = 20 m² b. ½ × 22 × 5 = 55 in²

EXIT TICKET

1. a. ½ × 14 × 4 = 28 cm² b. ½ × 26 × 3 = 39 ft²

2. ½ × (12 + 20) × 6 = ½ × 32 × 6 = 96 ft²

3. **C** (½ × (5 + 13) × 4)

PRACTICE FORM A

1. a. ½ × 24 × 8 = 96 cm² b. ½ × 48 × 15 = 360 in² c. ½ × 30 × 4 = 60 ft² d. ½ × 16 × 5 = 40 m²

2. a. 14 + 26 = 40 mi b. ½ × 40 × 10 = 200 mi² c. 40 mi

3. ½ × (18 + 30) × 11 = ½ × 48 × 11 = 264 in²

4. ½ × (8 + 20) × 7 = 98 m²

5. No. ½ × (11 + 17) × 20 = 280 cm². Talia forgot to take half.

6. ½ × (16 + 22) × 10 = 190 ft²; 190 × 2 = $380

7. M: ½ × 36 × 8 = 144 cm²; N: ½ × 36 × 8 = 144 cm². Same area, because the bases have the same sum (36) and the heights match.

8. a. 2 × 84 ÷ 7 = 24 m b. 24 − 10 = 14 m

9. Sofia used only one base. The formula adds both bases first: A = ½ × (7 + 21) × 5 = ½ × 28 × 5 = 70 m².

10. **B** (48 cm²)

11. **C** (add the bases)

Self-check word **FORMULA**

PRACTICE FORM B

1. a. ½ × 20 × 6 = 60 cm² b. ½ × 36 × 10 = 180 in² c. ½ × 44 × 5 = 110 ft² d. ½ × 16 × 9 = 72 m²

2. a. 18 + 30 = 48 mi b. ½ × 48 × 12 = 288 mi² c. 48 mi

3. ½ × (16 + 24) × 13 = ½ × 40 × 13 = 260 in²

4. ½ × (6 + 18) × 9 = 108 m²

5. No. ½ × (13 + 19) × 15 = 240 cm². Marcus forgot to take half.

6. ½ × (18 + 26) × 10 = 220 ft²; 220 × 3 = $660

7. M: ½ × 36 × 7 = 126 cm²; N: ½ × 36 × 7 = 126 cm². Same area, because the bases have the same sum (36) and the heights match.

8. a. 2 × 90 ÷ 6 = 30 m b. 30 − 11 = 19 m

9. Andre used only one base. The formula adds both bases first: A = ½ × (9 + 15) × 6 = ½ × 24 × 6 = 72 m².

10. **D** (56 in²)

11. **A** (two bases)

Self-check word **AVERAGE**

SHEET 1 · TRAPEZOIDS BY DECOMPOSING

1. a. $6 \times 4 = 24$ cm² b. triangle base $(10 - 6) \div 2 = 2$ cm; $\frac{1}{2} \times 2 \times 4 = 4$ cm² Area: $24 + 4 + 4 = 32$ cm²

2. Rectangle $7 \times 6 = 42$; triangle base $12 - 7 = 5$, $\frac{1}{2} \times 5 \times 6 = 15$; total 57 in²

3. Rectangle $3 \times 4 = 12$; triangle base $8 - 3 = 5$, $\frac{1}{2} \times 5 \times 4 = 10$; total 22 ft²

4. Rectangle $5 \times 3 = 15$; each triangle base $(11 - 5) \div 2 = 3$, $\frac{1}{2} \times 3 \times 3 = 4.5$; total $15 + 4.5 + 4.5 = 24$ ft²

5. The two triangle bases together equal $15 - 9 = 6$ cm, so each is 3 cm. Rectangle $9 \times 4 = 36$; triangles $2 \times (\frac{1}{2} \times 3 \times 4) = 12$; area = 48 cm².

SHEET 2 · THE TRAPEZOID AREA FORMULA

1. $\frac{1}{2} \times (8 + 14) \times 5 = 55$ m²

2. a. $\frac{1}{2} \times 14 \times 6 = 42$ ft² b. $\frac{1}{2} \times 14 \times 8 = 56$ in² c. $\frac{1}{2} \times 18 \times 3 = 27$ m² d. $\frac{1}{2} \times 10 \times 9 = 45$ cm²

3. $\frac{1}{2} \times (16 + 24) \times 10 = 200$ ft²; $200 \div 25 = 8$ bags

4. $2 \times 60 = 120$; $120 \div 6 = 20 = 8 + b$, so b = 12 in

5. **B** (40 cm²)

6. **C** ($\frac{1}{2} \times (3 + 9) \times 4$)

October 2026 conceptual additions — teacher key

Session 1 · Trapezoids by Decomposing

Area 32 cm². A valid decomposition may use a 5 × 4 rectangle and triangle areas totaling 12. The leftover bases sum to 6 but need not be equal unless the geometry supports it.

Extension: Accept a mathematically valid contrasting example and an explanation that uses the conditions in this session. Teacher anchor: Area 32 cm². A valid decomposition may use a 5 × 4 rectangle and triangle areas totaling 12. The leftover bases sum to 6 but need not be equal unless the geometry supports it.

Scoring: 1 point for a valid representation/setup, 1 for accurate mathematics and units, 1 for an explanation connecting the method to the task. Accept equivalent reasoning and suitable oral or diagram responses. Do not infer mastery from confidence alone.

Session 2 · The Trapezoid Area Formula

Combined base 20 m and height 5 m: area 100 m²; one trapezoid is 50 m². Formula (7+13) × 5 ÷ 2.

Extension: Accept a mathematically valid contrasting example and an explanation that uses the conditions in this session. Teacher anchor: Combined base 20 m and height 5 m: area 100 m²; one trapezoid is 50 m². Formula (7+13) × 5 ÷ 2.

Scoring: 1 point for a valid representation/setup, 1 for accurate mathematics and units, 1 for an explanation connecting the method to the task. Accept equivalent reasoning and suitable oral or diagram responses. Do not infer mastery from confidence alone.